Set up Hugging Face models

Every model ImageSkin downloads from Hugging Face, what it is for, and how to set up access from scratch on a laptop or on the hosted site. Source of `Set-up-Hugging-Face-Models.pdf`. Last checked against the code on 2026-10-27 (roadmap R160).

In short

- Only one model needs anything from you: **Pocket TTS** (`kyutai/pocket-tts`), whose weights that clone a voice are gated. You accept its terms once on Hugging Face and give the app a read token.
- Every other model is public. It downloads by itself the first time it is used, with no account.
- On the hosted site, the token goes in `deploy/.env` as `HF_TOKEN`. On a laptop, sign in once with `hf auth login`.
- Without a token, the app still works. It speaks the person's own voice with Chatterbox, which is about ten times slower, so a reply can pause for many seconds between sentences.

The models the app uses

Each model is downloaded once, the first time it is needed, and kept in the Hugging Face cache. On the hosted site the cache is `/data/huggingface`, inside the app's data volume, so it survives rebuilds. On a laptop it is `~/.cache/huggingface` unless `HF_HOME` says otherwise. Revisions are pinned in the code, so every machine gets the weights that were tested.

Model (Hugging Face repository)	What the app uses it for	Size	Access	Where in the code
`hexgrad/Kokoro-82M`	The ready-made voices (such as `af_heart`), the default people's voices, and the pronunciation step that splits words into sounds for the mouth	about 330 MB	Public	`kokoro_engine.py` (`REPO_ID`)
`kyutai/pocket-tts`	**Pocket TTS**: speaks in the person's own voice, cloned from 10 s of their voice sample, about ten times faster than Chatterbox (R22f, R160)	about 530 MB, with the next row	**Gated: accept the terms, then a read token**	`pocket_engine.py` (`TERMS_URL`)
`kyutai/pocket-tts-without-voice-cloning`	Pocket TTS's tokenizer, downloaded with the row above	(included above)	Public	the `pocket-tts` package
`ResembleAI/chatterbox-turbo` (revision `749d1c1a`)	**Chatterbox Turbo**: the person's own voice when Pocket TTS can't be used, and the converter that gives a voice another accent (American or British) once, in Prepare	about 3 GB with its libraries' models	Public	`chatterbox_engine.py` (`TURBO_REVISION`)
`facebook/wav2vec2-base-960h` (revision `22aad52d`)	Forced alignment: hears when each word of a cloned voice is said, so the mouth moves in time	about 360 MB	Public	`alignment.py` (`ALIGNER_REPO`)

Model (Hugging Face repository)	What the app uses it for	Size	Access	Where in the code
`KlingTeam/LivePortrait` (revision `82a4fa67`)	The photoreal face: the models that move the photo's head, eyes and mouth	about 520 MB	Public	`liveportrait.py` (`HF_REPO`)
`csukuangfj/speaker-embedding-models` (revision `0743f301`)	Speaker checks: whether the recordings are all the same person (one ONNX file)	about 30 MB	Public	`speaker_checks.py` (`MODEL_URL`)

The LLM

The local LLM is also downloaded from Hugging Face, by llama.cpp's `llama-server` rather than by the app. It is public and needs no token.

Model	Where	Size	Kept in
`unsloth/gemma-3-4b-it-GGUF`, quantization `Q4_0`	The hosted site (`deploy/compose.yaml`, the `llama` service)	about 2.4 GB	the `llama-models` volume (`LLAMA_CACHE=/models`)
`ggml-org/gemma-3-4b-it-GGUF`	A laptop (README, "Chat with the LLM")	about 2.5 GB	llama.cpp's own cache

Only for making the default people

The tools in `tools/default_people/` make the 60 default people once, on a machine with a graphics card. The app never downloads these.

Model	What it is for	Access
`segmind/SSD-1B` (revision `60987f37`)	Makes each default person's picture (`make_default_people.py`)	Public
`hexgrad/Kokoro-82M`	Lets you hear the voices to choose from (`hear_voices.py`)	Public

Not from Hugging Face

For a complete picture, these are downloaded from elsewhere and need no account:

- spaCy's English model `en_core_web_sm`, used by Kokoro, which is installed in the Docker image's build.
- MediaPipe's face landmarker and image segmenter, from Google's storage.
- LivePortrait's code, fetched from GitHub with git on the first Prepare.

Set up access: the hosted site

You do this once per Hugging Face account and once per site.

1. **Make an account.** Make a free account at https://huggingface.co/join, or sign in.

2. **Accept Pocket TTS's terms.** Open https://huggingface.co/kyutai/pocket-tts. Near the top there is a box about sharing your contact information to access the model; click its button (**Agree and access repository**). If the page already says "You have been granted access to this model", this is done.
3. **Make a read token.** Open https://huggingface.co/settings/tokens and click **Create new token**. Pick **Read**, give it any name and click **Create token**. Copy it: it starts with `hf_`. An existing token works too, if it is a Read token, or a fine-grained token with "Read access to contents of all public gated repos you can access" turned on.
4. **Put it in `.env`.** On the server, in `~/ImageSkinForLLM/deploy`, run `nano .env` and paste the token after `HF_TOKEN=`, such as `HF_TOKEN=hf_AbCd...`. Save with Ctrl+O and Enter, and close with Ctrl+X.
5. **Rebuild and restart:** `docker compose up -d --build`. The token reaches the app only when the container starts.
6. **Check it.** This prints `True` once the app has the token:

   ```
   docker compose exec app python -c "from huggingface_hub import get_token; print(get_token() is not None)"
   ```

7. **Prepare again.** On the Setup page, click **Prepare**. Then `docker compose logs app | grep "Prepare speaks"` should say `cloned with Pocket TTS`.

The `.env` file

`deploy/.env` holds the hosted site's settings and secrets. It stays on the server only: it is never committed, and `chmod 600 .env` makes it readable only by you. Start from the example with `cp .env.example .env`. It has four settings:

Setting	What it is	Example
`SITE_ADDRESS`	The site's name, whose A record points at the server. With no domain, the server's IP with dashes and `.sslip.io`.	`app.me0.ai`
`LOGIN_NAME`	The name the browser asks for.	`larry`
`LOGIN_PASSWORD_HASH`	The password's hash, from `docker run --rm -it caddy:2 caddy hash-password`, between single quotes.	`'$2a$14$...'`
`HF_TOKEN`	The Hugging Face read token, for Pocket TTS. Leave it empty to speak with Chatterbox.	`hf_AbCd...`

`deploy/compose.yaml` passes `HF_TOKEN` to the app's container; `huggingface_hub` reads it from there. A change to `.env` takes effect with `docker compose up -d` (add `--build` when the code changed too).

Set up access: a laptop

1. Do steps 1 to 3 above.
2. In the app's virtual environment, run `hf auth login` and paste the token. It is saved in the Hugging Face cache, so this is done once.
3. Install Pocket TTS as the README's "Replies within 2 seconds in your own voice" says, then start `imageskin serve` and click **Prepare**.

When something goes wrong

- **The log says `Pocket TTS is installed but not signed in to Hugging Face; speaking with Chatterbox.`** The app has no token. On the site, check that `HF_TOKEN` is set in `.env` and that the container was started again after; on a laptop, run `hf auth login`.
- **Prepare fails with `could not load Pocket TTS: its weights that clone a voice need the terms accepted.`** The token works, but the terms on https://huggingface.co/kyutai/pocket-tts were not accepted for that account, or the token can't read gated models. The message ends with what Hugging Face said.
- **A download is slow or stops.** Each model downloads once, then is kept. A stopped download carries on the next time. `docker compose logs -f app` shows progress.
- **To start over** with the models (rarely needed), remove the cache folder `/data/huggingface` inside the app's volume, or `~/.cache/huggingface` on a laptop. They download again on next use.